

Self-triggered Sampling Design For Continuous-time Linear Dynamical System

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Abstract—A novel self-triggered mechanism is developed in this paper for the regulation problem of continuous-time linear systems to improve computational efficiency and reduce communication load. In contrast to conventional event-triggered control strategies that needs constant state monitoring, this work develops a self-triggered sampling rule that determines state update instants in advance without requiring continuous state measurement. With Zeno-free property guaranteed, the triggering condition is analytically constructed using upper and lower bounds of the state and sampling error, respectively. By formulating the control objective through a quadratic performance index, the optimal control input is derived by solving the algebraic Riccati equation (ARE) with intermittent state feedback. The asymptotic stability of the closed-loop system is verified by theoretical analysis and simulation example under the presented self-triggered optimal control scheme, while notably decreasing the frequency of feedback updates.

Index Terms—Self-triggered mechanism, Event-triggered control, ARE, Optimal control.

I. INTRODUCTION

Optimal control theory serves as a foundational concept in control engineering for dynamical systems [1]–[3]. The central objective is to design control policies that optimize a given performance criterion while ensuring system stability [4], [5]. For continuous-time linear systems, the linear quadratic regulator (LQR) framework is a classical and widely adopted control strategy [6]–[8]. By minimizing a quadratic performance criterion that incorporates the system state and control input, the LQR framework yields an optimal control law based on the solution to the ARE [7]. This control law, when implemented with continuous state feedback, guarantees asymptotic stability and optimal performance, making it a widely adopted approach in both theory and applications [9].

The practical implementation of continuous-time optimal controllers faces limitations in systems constrained by computational or communication resources [10]. Continuous state monitoring and frequent actuation updates are costly and often impractical for embedded or networked environments. To address this, alternative control strategies have been developed to

reduce unnecessary updates without sacrificing performance. Event-triggered control (ETC) alleviates these demands by updating the control input only when a state-dependent triggering condition is violated, rather than at fixed intervals [11]. Consequently, the ETC strategy has obtained substantial attention from researchers in the control field. Wang *et al.* investigated a model predictive control scheme with event-triggered mechanisms for power converters, aiming to reduce switching losses while still maintaining effective regulation performance [12]. Zhang *et al.* developed an adaptive resilient ETC strategy for rear-wheel-drive autonomous vehicles, which effectively manages the frequency and magnitude of control adjustments during vehicle operation [13]. For the nonlinear multi-agent system under an intermittent ETC mechanism, Liu *et al.* investigated an optimal leader-following consensus approach with convergence and Zeno-free property guaranteed [14]. Despite these successful applications, there still exists notable drawbacks of ETC, which requires constant state monitoring and can be computationally expensive.

This motivates the development of self-triggered control (STC), where future control update instants are computed offline based on current state samples [15]–[17]. STC avoids continuous monitoring while preserving control performance, thereby enabling broader utilization in both practical implementations and theoretical research. For nonlinear discrete-time systems, Sun *et al.* investigated a robust STC strategy considering additive disturbances and constraints, which resulted in reduced sampling frequency and upper-bounded performance loss [18]. Liu *et al.* combined the STC strategy with a data-driven method that calculates a series of control inputs alongside the prediction of the system's trajectory [19]. For DC micro-grids, Peng *et al.* applied the discrete-time STC under data dropouts and communication delays while proactively estimating the successive update instants ahead of time [20]. However, these above-mentioned results do not take optimal control performance into account, which limits the applicability. This limitation serves as the main motivation for our work in this paper, where we aim to integrate optimality into the control design while maintaining the advantages of self-triggered strategy.

The main contribution of this paper can be collected as

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follows:

- Unlike current ETC mechanisms [11], [21] that requires continuous state monitoring, the proposed method eliminates this drawback by analytically constructing a self-triggering condition. This enables the offline prediction of successive triggering instants, achieving reduced communication burden while excluding the Zeno phenomenon.
- In contrast to prior self-triggered schemes [15], [22] that often lack the integration with optimal control objectives, this work explicitly couples the self-triggered sampling mechanism with an optimal control input derived from the ARE. The control input under intermittent feedback asymptotically stabilizes the closed-loop system while minimizing the quadratic performance index, thus bridging the gap between optimality and efficient resource usage.

II. PROBLEM STATEMENT

This paper focuses on the continuous-time linear system, which can be formulated as

$$\dot{s}(t) = As(t) + Bu(t), \quad s(0) = s_0, \quad (1)$$

where $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$ correspond to the state and input matrix, respectively, $u(t) \in \mathbb{R}^m$ denotes the control input, and $s(t) \in \mathbb{R}^n$ is the system state.

The following performance index is utilized to characterize the regulation behavior of system (1), as given by

$$\mathcal{V}(u; s_0) \triangleq \int_0^\infty h(s(\tau), u(\tau)) d\tau \quad (2)$$

where the reward function is given by $h(s, u) \triangleq \frac{1}{2}(\|s\|_Q^2 + \|u\|_R^2)$ with $\|s\|_Q^2 \triangleq s^\top Q s$ and $\|u\|_R^2 \triangleq u^\top R u$ for $Q \succeq 0$ and $R \succ 0$ being symmetric matrices. The following assumption and definition are related to subsequent discussions.

Definition 1. [23] *The feedback control policy $u(s)$ is admissible with respect to the performance index (2), denoted as $u \in \Psi(\Omega)$, where Ω is a compact set, if the following conditions hold:*

- $u(s)$ is continuous;
- $\mathcal{V}(u; s_0)$ is finite $\forall s_0 \in \Omega$;
- $u(0) = 0$;
- $u(s)$ stabilizes the system (1).

Assumption 1. [24] *The pair (A, B) is stabilizable and the pair $(A, \sqrt{Q})$ is detectable.*

Considering that the control input is admissible, the Hamiltonian function can be denoted as

$$\begin{aligned} \mathcal{H}(s, u(s), \nabla V(s)) \\ = \langle \nabla V(s), \dot{s} \rangle + h(s, u) \\ = \nabla^\top V(s) [As + Bu(s)] + \frac{1}{2}(\|s\|_Q^2 + \|u\|_R^2). \end{aligned} \quad (3)$$

Under Assumption 1, $V(s)$ is the value function that uniquely solves the Hamiltonian-Jacobi-Bellman (HJB) equation [25]

$$\mathcal{H}(s, u(s), \nabla V(s)) = 0. \quad (4)$$

Based on the optimal control theory [2], [3], the optimal value function can be determined as

$$V^*(s) \triangleq \min_{u \in \Psi(\Omega)} \int_0^\infty h(s(\tau), u(\tau)) d\tau.$$

Given the linear dynamics described in (1) and the quadratic structure of the reward $h(s, u)$, $V^*(s)$ defined in (5) takes a quadratic form as

$$V^*(s) = \frac{1}{2} s^\top P s, \quad (5)$$

with the symmetric matrix $P \succ 0$. Enforcing the stationary condition $\partial \mathcal{H}(s, u(s), \nabla V^*(s)) / \partial u = 0$, the control input can be derived in an optimal manner, as given by

$$\begin{aligned} u^*(s) &= \arg \min_{u \in \Psi(\Omega)} \mathcal{H}(s, u(s), \nabla V^*(s)) \\ &= -R^{-1} B^\top P s. \end{aligned} \quad (6)$$

The following standard assumptions are made on $u^*(s)$ and the system (1).

Assumption 2. [26] *With $L > 0$, the optimal control law $u^*(s)$ satisfies the Lipschitz continuity condition on the compact set Ω , i.e.,*

$$\|u^*(p) - u^*(q)\| \leq L \|p - q\|, \quad \forall p, q \in \Omega. \quad (7)$$

Assumption 3. [16] *For any $s, \hat{s} \in \Omega$ and $\tilde{s} = s - \hat{s}$, $\exists \ell > 0$ such that*

$$\begin{aligned} \langle s, As + Bu^*(\hat{s}) \rangle &\leq \ell (1 + \|s\|^2 + \|\tilde{s}\|), \\ \langle \tilde{s}, As + Bu^*(\hat{s}) \rangle &\leq \ell (1 + \|s\|^2 + \|\tilde{s}\|). \end{aligned} \quad (8)$$

By inserting (5) and (6) into the HJB equation (4), the ARE can be determined as follows,

$$Q + A^\top P + PA - PBR^{-1}B^\top P = 0, \quad (9)$$

which admits a unique solution in terms of the matrix P .

Note that the control law $u^*(s)$ presented in (6) requires continuous state feedback, which may lead to high computational demand and increased communication overhead. To mitigate this challenge, we introduce the concept of triggered optimal control, wherein $u^*(s)$ is implemented under a triggering mechanism, as elaborated in the subsequent section.

III. EVENT-TRIGGERED SAMPLING

In this paper, the central objective is to develop a sampling strategy applicable to the system state $s(t)$. Therefore, two kinds of sampled-feedback control, namely ETC and STC approaches, are explored in detail as follows.

A. Event-triggered framework

To enhance computational efficiency and alleviate communication load, an ETC method is developed based on an state sampling strategy given as

$$\hat{s}(t) = \begin{cases} s(t_i) & , t \in [t_i, t_{i+1}) \\ s(t_{i+1}) & , t = t_{i+1} \end{cases} \quad (10)$$

for an increasing sequence of sample instants $\langle t_i \rangle$, $i \in \mathbb{N}$. The corresponding sampling error $\tilde{s}(t)$ is given by $\tilde{s}(t) \triangleq s(t) - \hat{s}(t)$. Therefore, the event-based control input under sample process is determined as

$$u^*(\hat{s}) = -R^{-1}B^\top P\hat{s}. \quad (11)$$

The algebraic property of the Hamiltonian $\mathcal{H}$ with intermittent sampled state $\hat{s}$ in $u^*(\hat{s})$ differs from that under continuous feedback, as described in the following lemma.

Lemma 1. [26] *The event-based Hamiltonian satisfies*

$$\mathcal{H}(s, u^*(\hat{s}), \nabla V^*(s)) = \|u^*(s) - u^*(\hat{s})\|_R^2. \quad (12)$$

Meanwhile, the sample process leads to the degraded performance, as discussed in the next lemma.

Lemma 2. [15] *Suppose Assumptions 1 and 2 hold. With a positive constant $\alpha \in (0, 1)$, the sample-based control input $u^*(\hat{s})$ in (11), and the sample process (10), the event-triggering condition is established as*

$$\|\tilde{s}\|^2 \leq \frac{\lambda_{\min}(Q)(1-\alpha)\|s\|^2 + \|u^*(\hat{s})\|_R^2}{L^2\lambda_{\max}(R)}. \quad (13)$$

Then, the following propositions can be obtained:

- 1) the value function V with event-triggered rule satisfies

$$\begin{aligned} & V(s_0; u^*(\hat{s})) - V(s_0; u^*(s)) \\ &= \int_0^\infty \|u^*(s(\tau)) - u^*(\hat{s}(\tau))\|_R^2 d\tau, \end{aligned} \quad (14)$$

- 2) the closed-loop system (1) is stabilized asymptotically.

B. Self-triggered framework

In conventional event-triggering condition (13), the sample rule needs continuous state observation, which brings computational overload and increases communication burden. To completely avoid this issue, the self-triggered mechanism is proposed that no longer requires continuous state monitoring, and determines the trigger instants in advance. For this purpose, we firstly aim to remove the state $s(t)$ from the trigger condition. The following lemma offers such an approach, which relates the terms $\|\tilde{s}\|^2$ and $\|s\|^2$ in (13) to the sampled state $\hat{s}$.

Lemma 3. *Under Assumption 3 and the sample process in (10), consider the linear system (1). Then, for all $t \in [t_i, t_{i+1})$, $i \in \mathbb{N}$, $\|\tilde{s}\|^2$ and $\|s\|^2$ satisfies*

$$\|\tilde{s}(t)\|^2 \leq \theta_1(\|\hat{s}\|, t - t_i), \quad (15)$$

$$\|s(t)\|^2 \geq \theta_2(\|\hat{s}\|, t - t_i), \quad (16)$$

where

$$\begin{aligned} \theta_1(\|\hat{s}\|, t - t_i) &= \frac{2\|\hat{s}\|^2 + 1}{3} (e^{6\ell t} - 1), \\ \theta_2(\|\hat{s}\|, t - t_i) &= \frac{5\|\hat{s}\|^2 + 1}{3} e^{-6\ell t} - \frac{2\|\hat{s}\|^2 + 1}{3}. \end{aligned}$$

Proof. Recall (11). The dynamics of the state and the sample error can be specified respectively as

$$\dot{s}(t) = As(t) + Bu^*(\hat{s}), \quad (17)$$

$$\dot{\tilde{s}}(t) = -[As(t) + Bu^*(\hat{s})]. \quad (18)$$

Then, with Assumption 3, differentiating $\|\tilde{s}\|^2$ yields

$$\begin{aligned} \frac{d}{dt}\|\tilde{s}\|^2 &= -2\langle \tilde{s}, As + Bu^*(\hat{s}) \rangle \leq 2\ell(1 + \|s\|^2 + \|\tilde{s}\|^2) \\ &= 2\ell(1 + \|\hat{s} + \tilde{s}\|^2 + \|\tilde{s}\|^2) \\ &\leq 2\ell(1 + 2\|\hat{s}\|^2 + 3\|\tilde{s}\|^2). \end{aligned} \quad (19)$$

On the time interval $[t_i, t)$, applying the Comparison Lemma [27] on (19), one has

$$\begin{aligned} & \|\tilde{s}(t)\|^2 \\ & \leq 2\ell(1 + 2\|\hat{s}\|^2) \int_{t_i}^t e^{6\ell(\tau - t_i)} d\tau + e^{6\ell(t - t_i)} \|\tilde{s}(t_i)\|^2 \\ & = 2\ell(1 + 2\|\hat{s}\|^2) \int_{t_i}^t e^{6\ell(\tau - t_i)} d\tau. \quad (\text{By } \tilde{s}(t_i) = 0) \\ & = \theta_1(\|\hat{s}\|, t - t_i). \end{aligned} \quad (20)$$

Then, taking time derivative of $\|s\|^2$, one has

$$\frac{d}{dt}(-\|s(t)\|^2) = -2\langle s, As + Bu^*(\hat{s}) \rangle. \quad (21)$$

Using Assumption 3 and following the similar derivation of (19) - (20), one can obtain (16). $\square$

In the following theorem, with the inequalities in (15) and (16), we present the principle results of this paper.

Theorem 1. *Suppose Assumption 3 holds under the sample process (10). With constants $\gamma_1, \gamma_2 > 0$, $\rho_1 \in \mathcal{K}$ being convex and $\rho_2 \in \mathcal{K}_\infty$ satisfying $\rho_1(\chi) \leq \chi \leq \rho_2(\chi)$, $\forall \chi \in [0, \infty)$, the self-triggered rule formulated as*

$$\rho_2(\theta_1(\|\hat{s}\|, t - t_i)) \leq \gamma_1 \rho_1(\theta_1(\|\hat{s}\|, t - t_i) + \gamma_2) \quad (22)$$

guarantees the Zeno-free property.

Proof. We begin by showing that the threshold equation $\Xi(t; t_i) = 0$ admits a unique solution, where

$$\Xi(t; t_i) \triangleq \rho_2(\theta_1(\|\hat{x}\|, t - t_i)) - \gamma_1 \rho_1(\theta_1(\|\hat{x}\|, t - t_i) + \gamma_2).$$

Given that θ_1 and θ_2 being increasing and decreasing in t , and ρ_1, ρ_2 satisfy the stated properties, it follows that $\Xi(t; t_i)$ is monotonically increasing in t . Observing that $\Xi(t_i; t_i) < 0$ and $\Xi(t; t_i) > 0$ as $t \rightarrow \infty$, intermediate value theorem ensures that a unique solution exists in the interval $[t_i, \infty)$. This solution specifies the next triggering instant.

The interval $\Delta t_i \triangleq t_{i+1} - t_i$ is shown to be lower-bounded by a positive constant to prevent the Zeno phenomenon. To this end, there always exists a function such that $0 < \varpi(\|\hat{s}\|) < \|\hat{s}\|^2 + \gamma_2$. Inspired by [16], consider a sufficient condition

$$\theta_2(\|\hat{s}\|, t - t_i) + \gamma_2 \leq \varpi(\|\hat{s}\|) \leq \theta_1(\|\hat{s}\|, t - t_i). \quad (23)$$

Solving above inequality, one can obtain $\Delta t_i \geq T_i > 0$, where

$$\begin{aligned} T_i &\triangleq \frac{1}{6\ell} \min \{ \delta_1 (\|\hat{s}\|), \delta_2 (\|\hat{s}\|) \}, \\ \delta_1 (\|\hat{s}\|) &\triangleq \ln \left(1 + \frac{3\varpi (\|\hat{s}\|)}{1 + 2\|\hat{s}\|^2} \right), \\ \delta_2 (\|\hat{s}\|) &\triangleq -\ln \left(1 - \frac{3(\gamma_2 + \|\hat{s}\|^2 - \varpi (\|\hat{s}\|))}{1 + 5\|\hat{s}\|^2} \right). \end{aligned}$$

This completes the proof. $\square$

In this paper, for the presented self-triggered mechanism (22), we set $\rho_1(\chi) = \rho_2(\chi) = \chi$. Therefore, rewriting the threshold equation, one can obtain

$$\Xi(t; t_i) = \theta_1(\|\hat{s}\|, t - t_i) - \gamma_1 \theta_1(\|\hat{s}\|, t - t_i) - \gamma_1 \gamma_2 = 0.$$

Under the proposed self-triggered strategy and the optimal control input $u^*(\hat{s})$ with intermittent feedback, the property of system described in (1) can be formalized in the following.

Theorem 2. *Let Assumptions Assumptions 1 to 3 hold. Consider the sample process (10) and the self-triggered rule (22). Then, the event-based optimal control input $u^*(\hat{s})$ in (11) asymptotically stabilizes the system (1).*

Proof. The optimal value function $V^*(s)$ defined in (5) can be determined as the Lyapunov candidate due to the positive definiteness. Hence, differentiating $V^*(s)$, one has

$$\frac{d}{dt} V^*(s) = \dot{V}^*(s) = \nabla^\top V^* A s - \nabla^\top V^* B R^{-1} B^\top P \hat{s}.$$

Recall (3) and (4). The term $\nabla^\top V^* A s$ satisfies

$$\nabla^\top V^*(s) A s = -\nabla^\top V^*(s) B u^*(s) - h(s, u^*(s)). \quad (24)$$

Substituting (6) and $\nabla^\top V^*(s) = s^\top P$ into (24) yields

$$\nabla^\top V^*(s) A s = \frac{1}{2} s^\top P B R^{-1} B^\top P s - \frac{1}{2} \|s\|_Q^2,$$

which can be inserted into (24). Then, $\dot{V}^*(s)$ becomes

$$\begin{aligned} \dot{V}^*(s) &= -\frac{1}{2} \|s\|_Q^2 + \frac{1}{2} \|u^*(s)\|_R^2 - u^*(s)^\top R u^*(\hat{s}) \\ &= -\frac{1}{2} \|s\|_Q^2 - \frac{1}{2} \|u^*(\hat{s})\|_R^2 + \frac{1}{2} \|u^*(s) - u^*(\hat{s})\|_R^2 \end{aligned}$$

Applying Assumption 2 to above equation, $\dot{V}^*$ further satisfies

$$\begin{aligned} \dot{V}^*(s) &\leq -\frac{\lambda_{\min}(Q)}{2} \|s\|^2 - \frac{1}{2} \|u^*(\hat{s})\|_R^2 + \frac{L^2 \lambda_{\max}(R)}{2} \|\tilde{s}\|^2 \\ &= -\frac{1}{2} \lambda_{\min}(Q) \sigma \|s\|^2 - \frac{1}{2} \|u^*(\hat{s})\|_R^2 \\ &\quad - \frac{1}{2} \lambda_{\min}(Q) \bar{\sigma} \|s\|^2 + \frac{1}{2} L^2 \lambda_{\max}(R) \|\tilde{s}\|^2, \end{aligned} \quad (25)$$

where $\sigma \in (0, 1)$ and $\bar{\sigma} = 1 - \sigma$.

Next, we need to remove the term $\|\tilde{s}\|$ in (25). In event-triggered mechanism, Lemma 2 plays a role. While in this paper, we implement the self-triggering condition (22) to establish the relation between $\|\tilde{s}\|$ and $\|s\|$. Specifically, since

$\rho_1 \in \mathcal{K}$ and $\rho_2 \in \mathcal{K}_\infty$, substituting the inequalities (15) and (16) into (22), one has

$$\|\tilde{s}\|^2 \leq \gamma_1 \|s\|^2 + \gamma_1 \gamma_2. \quad (26)$$

Hence, by selecting the parameters such that $\gamma_1 = \frac{\lambda_{\min}(Q) \bar{\sigma}}{L^2 \lambda_{\max}(R)}$ and $\gamma_2 = \frac{\|u^*(\hat{s})\|_R^2}{\lambda_{\min}(Q) \bar{\sigma}}$, (26) further satisfies

$$\|\tilde{s}\|^2 \leq \frac{\lambda_{\min}(Q) \bar{\sigma} \|s\|^2 + \|u^*(\hat{s})\|_R^2}{L^2 \lambda_{\max}(R)}. \quad (27)$$

Finally, inserting (27) into (25) yields

$$\dot{V}^*(s) \leq -\frac{1}{2} \lambda_{\min}(Q) \sigma \|s\|^2 \leq 0, \quad (28)$$

which shows the asymptotically stability of the system (1) at the origin. This complete the proof. $\square$

From above derivation, one can observe that (22) is a sufficient condition to (13), and this indicates the condition of STC is more conservative than that of ETC.

IV. SIMULATION STUDY

To investigate the effectiveness of presented the self-triggered optimal control strategy, the F16 aircraft model described in [2] is considered in this section, as given by:

$$\dot{s}(t) = \begin{bmatrix} -1.01887 & 0.90506 & -0.00215 \\ 0.82225 & -1.07741 & -0.17555 \\ 0 & 0 & -1 \end{bmatrix} s + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u,$$

where $s = [s_1, s_2, s_3]^\top$ is the state. The initial condition is given by $s(0) = [0.1, 0.2, 0.3]^\top$. The parameters Q and R in the performance index (2) are selected as $Q = 0.1I$ and $R = 1$. By solving the ARE in (9), one has

$$P = \begin{bmatrix} 0.1468 & 0.1213 & -0.0184 \\ 0.1213 & 0.1481 & -0.0202 \\ -0.0184 & -0.0202 & 0.0522 \end{bmatrix}, \quad (29)$$

by which the optimal control with intermittent feedback is formulated as

$$u^*(\hat{s}) = -0.0184 \hat{s}_1 - 0.0202 \hat{s}_2 + 0.0522 \hat{s}_3.$$

Under the self-triggered mechanism in (22) with the parameters chosen as $\ell = 0.1$, $L = 0.8$ and $\sigma = 0.5$, the simulation results can be obtained as depicted in Figures 1 to 4.

Fig. 1 illustrates the evolution of the state $s(t)$, along with their corresponding sampled states $\hat{s}(t)$ and the sampling error $\tilde{s}(t)$. The results demonstrate that, despite the aperiodic sample process, the states converge asymptotically to the origin, confirming system stability under self-triggered control. The errors grow between sampling intervals and reset at trigger instants, validating the boundedness guaranteed by Lemma 3. This shows that the triggering instants are determined by (22) to ensure performance while minimizing unnecessary updates.

The optimal control input trajectories with continuous state feedback and intermittent (self-triggered) feedback are presented in Fig. 2. The control signal under self-triggering follows a piecewise constant pattern, which corresponds to

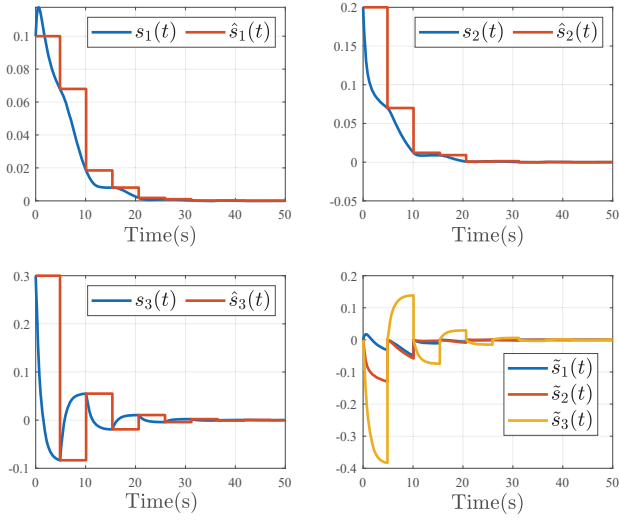


Fig. 1. Evolution of the system state $s_1(t)$, $s_2(t)$, $s_3(t)$ and sampled state with sample errors.

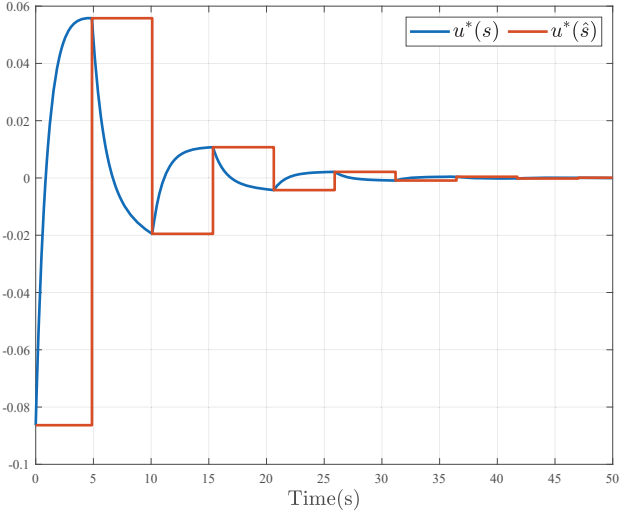


Fig. 2. Evolution of the control input with continuous/intermittent feedback.

the evolution of sampled state as shown in Fig. 1. Despite the sparse updates, the control input remains close to that under the continuous-feedback, highlighting the ability of the presented STC scheme to preserve control performance with reduced resource usage. The evolution of the instantaneous cost $h(s, u)$ and the performance index are depicted in Fig. 3. The convergence of h validates the theoretical analysis in Theorem 2. From Fig. 4, it is clear that the STC significantly decreases the update frequency, thereby reduce more communication bandwidth than ETC approach.

V. CONCLUSION

This paper presents a novel optimal STC mechanism for a regulation problem in continuous-time linear systems. Through the upper and lower bounds of the sample error and state,

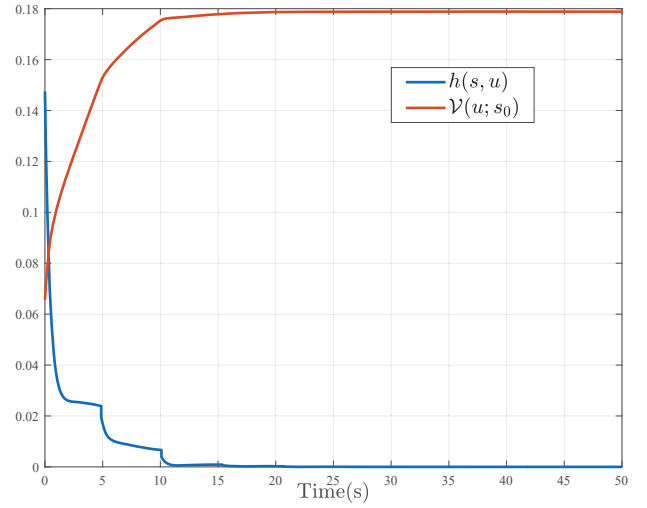


Fig. 3. Evolution of the reward $h(s, u)$ and the performance index.

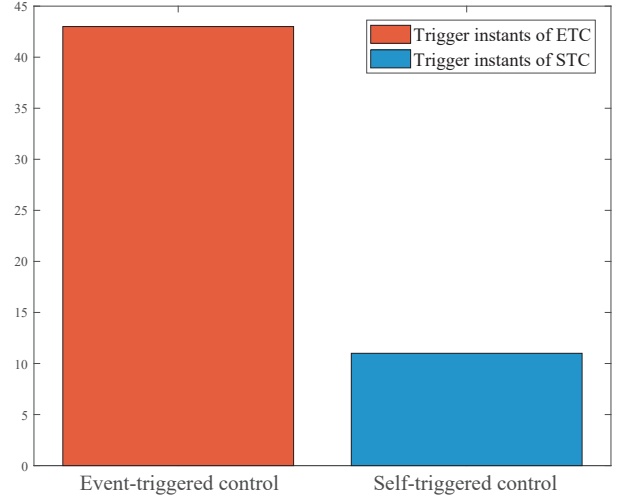


Fig. 4. Trigger instants under ETC and STC.

respectively, a self-triggering condition is constructed which eliminates the need for continuous state monitoring and guarantees the Zeno-free property. Within the self-triggered sampling structure, the optimal control input with intermittent state feedback guarantees asymptotic stability and determines the successive sampling instants, while overcoming the limitation of conventional event-triggered approach by significantly reducing the computational load. Theoretical discussions and simulation example validate the efficiency of the self-triggered optimal control method developed in this paper. Future works will extended the results to multi-agent systems and dynamical game.

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